

Math 1512 Calculus 1 Review: Selected Solutions

Fall 2026

Problem numbers match the review sheet. Each problem is restated before its solution.

1. Formulas to know.

- (a) State the definition of f is continuous at $x = a$.

Solution. f is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

This one equation hides three requirements: $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and the two are equal. A hole fails the third, a jump fails the second, a vertical asymptote fails all three. ■

- (b) Write $f'(a)$, the derivative of f at $x = a$, as a limit in two ways.

Solution.

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

Both are limits of slopes of secant lines through $(a, f(a))$. The second form just names the horizontal gap $h = x - a$. ■

- (l) Give both parts of the Fundamental Theorem of Calculus.

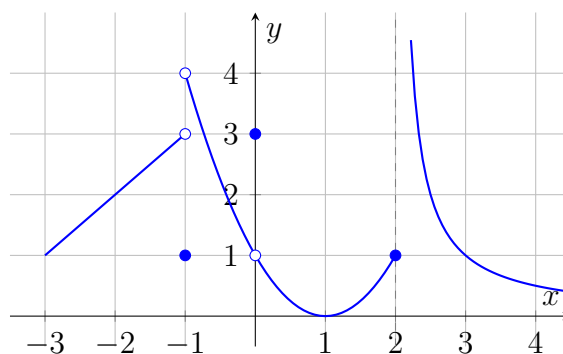
Solution. Let f be continuous on $[a, b]$.

Part 1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$. The rate the area grows is the height of the graph at the moving edge.

Part 2. If F is any antiderivative of f , meaning $F' = f$, then $\int_a^b f(x) dx = F(b) - F(a)$. Adding up tiny changes in F gives the total change in F . ■

2. Limits from a graph.

- (a-c) Let g be the function graphed below (a filled dot is a point on the graph, an open dot is not). Find $g(-1)$, $\lim_{x \rightarrow -1^-} g(x)$, $\lim_{x \rightarrow -1^+} g(x)$, $\lim_{x \rightarrow -1} g(x)$, then $g(0)$ and $\lim_{x \rightarrow 0} g(x)$, then $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$.



Solution. At $x = -1$ the filled dot gives $g(-1) = 1$. From the left the line climbs to the open dot at height 3, from the right the parabola comes down from the open dot at height 4.

$$\lim_{x \rightarrow -1^-} g(x) = 3, \quad \lim_{x \rightarrow -1^+} g(x) = 4,$$

so $\lim_{x \rightarrow -1} g(x)$ does not exist, since the one-sided limits disagree.

At $x = 0$ the dot sits at $g(0) = 3$, but the curve on both sides heads to the open dot at height 1. So $\lim_{x \rightarrow 0} g(x) = 1$. The value of g at 0 has no effect on the limit.

At $x = 2$ the parabola arrives at the filled dot, so $\lim_{x \rightarrow 2^-} g(x) = 1$. To the right the graph shoots up along the dashed asymptote, so $\lim_{x \rightarrow 2^+} g(x) = \infty$. The two-sided limit does not exist. ■

- (d-e) At which x values in $(-3, 4)$ is g not continuous? Classify each discontinuity. At which of these is g continuous from the left, or from the right?

Solution. g is not continuous at $x = -1, 0, 2$.

At $x = -1$ the one-sided limits are 3 and 4: a jump. At $x = 0$ the limit exists ($= 1$) but is not $g(0) = 3$: removable, since redefining one point fixes it. At $x = 2$ the right-hand limit is infinite: an infinite discontinuity.

Continuous from the left at a means $\lim_{x \rightarrow a^-} g(x) = g(a)$. This holds only at $x = 2$, where both equal 1. No point is continuous from the right: at -1 we compare 4 with 1, at 0 we compare 1 with 3, and at 2 the right limit is infinite. ■

- (f) Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} h(x) = -2$. Evaluate $\lim_{x \rightarrow a} [3f(x) - 5h(x)]$, $\lim_{x \rightarrow a} f(x)h(x)$, $\lim_{x \rightarrow a} \frac{f(x)}{h(x)}$, and $\lim_{x \rightarrow a} \left(\sqrt{f(x)} + h(x)^2 \right)$.

Solution. The limit laws say limits pass through sums, constant multiples, products, quotients (with nonzero bottom limit), powers and roots. So we can replace $f(x)$ by 4 and $h(x)$ by -2 .

$$3(4) - 5(-2) = 22, \quad 4(-2) = -8, \quad \frac{4}{-2} = -2, \quad \sqrt{4} + (-2)^2 = 6.$$

3. Computing limits.

- (a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$.

Solution. Plugging in gives $\frac{0}{0}$, so $x - 3$ divides both.

$$\lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{(x-3)(x+3)} = \lim_{x \rightarrow 3} \frac{x+1}{x+3} = \frac{4}{6} = \frac{2}{3}.$$

Canceling is allowed because the limit never uses $x = 3$ itself. ■

- (b) Evaluate $\lim_{x \rightarrow 5} \frac{\sqrt{x+4}-3}{x-5}$.

Solution. $\frac{0}{0}$ again, and the root is the problem. Multiply top and bottom by the conjugate $\sqrt{x+4}+3$.

$$\frac{(x+4)-9}{(x-5)(\sqrt{x+4}+3)} = \frac{1}{\sqrt{x+4}+3} \longrightarrow \frac{1}{6}.$$

■

(c) Evaluate $\lim_{x \rightarrow -1} \frac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$.

Solution. Combine the fractions in the numerator over the common denominator $2(x+3)$.

$$\frac{1}{x+1} \cdot \frac{2-(x+3)}{2(x+3)} = \frac{1}{x+1} \cdot \frac{-(x+1)}{2(x+3)} = \frac{-1}{2(x+3)} \longrightarrow -\frac{1}{4}.$$

■

(d) Evaluate $\lim_{x \rightarrow 4} \frac{5}{x^2-3x-4}$, considering the one-sided limits.

Solution. The top is 5 and the bottom $(x-4)(x+1)$ goes to 0, so each one-sided limit is $\pm\infty$. Near $x=4$ the factor $x+1$ is near 5, positive, so the sign is the sign of $x-4$.

$$\lim_{x \rightarrow 4^-} \frac{5}{(x-4)(x+1)} = -\infty, \quad \lim_{x \rightarrow 4^+} \frac{5}{(x-4)(x+1)} = \infty.$$

The one-sided limits disagree, so the limit does not exist.

■

(f) Evaluate $\lim_{x \rightarrow 2} \frac{x^2-4}{|x-2|}$.

Solution. An absolute value is a piecewise function, so split into the two sides. For $x > 2$, $|x-2| = x-2$, and for $x < 2$, $|x-2| = -(x-2)$.

$$\lim_{x \rightarrow 2^+} \frac{(x-2)(x+2)}{x-2} = 4, \quad \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{-(x-2)} = -4.$$

The one-sided limits disagree, so the limit does not exist. The graph has a jump at $x=2$.

■

4. Squeeze Theorem.

(a) Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$.

Solution. $\sin\left(\frac{3}{x}\right)$ oscillates and has no limit, so the product law does not apply (this answers (c)). But it is bounded: $-1 \leq \sin\left(\frac{3}{x}\right) \leq 1$. Multiplying by $x^2 \geq 0$,

$$-x^2 \leq x^2 \sin\left(\frac{3}{x}\right) \leq x^2.$$

Both bounds go to 0, so by the Squeeze Theorem the limit is 0. The x^2 crushes the oscillation.

■

(b) If $2x-1 \leq f(x) \leq x^2-2x+3$ for all x , find $\lim_{x \rightarrow 2} f(x)$.

Solution. $\lim_{x \rightarrow 2} (2x - 1) = 3$ and $\lim_{x \rightarrow 2} (x^2 - 2x + 3) = 3$. Since f is squeezed between them, $\lim_{x \rightarrow 2} f(x) = 3$.

The bounds touch at $x = 2$ because $(x^2 - 2x + 3) - (2x - 1) = (x - 2)^2$. ■

5. Limits at infinity.

(b) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+1}}{x+3}$.

Solution. Pull x^2 out of the root: $\sqrt{4x^2+1} = \sqrt{x^2} \sqrt{4 + \frac{1}{x^2}} = |x| \sqrt{4 + \frac{1}{x^2}}$. For $x < 0$, $|x| = -x$.

$$\frac{-x \sqrt{4 + \frac{1}{x^2}}}{x \left(1 + \frac{3}{x}\right)} = \frac{-\sqrt{4 + \frac{1}{x^2}}}{1 + \frac{3}{x}} \rightarrow -2.$$

The sign makes sense: the top is positive and the bottom is negative for very negative x . ■

(c) Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$.

Solution. This is $\infty - \infty$, which is not determined yet. Multiply by the conjugate over itself.

$$\frac{(x^2 + 6x) - x^2}{\sqrt{x^2 + 6x} + x} = \frac{6x}{x \left(\sqrt{1 + \frac{6}{x}} + 1\right)} = \frac{6}{\sqrt{1 + \frac{6}{x}} + 1} \rightarrow \frac{6}{2} = 3.$$

■

(d) Find all vertical and horizontal asymptotes of $f(x) = \frac{2x^2-2}{x^2+x-2}$. Why is $x = 1$ not a vertical asymptote?

Solution. Factor: $f(x) = \frac{2(x-1)(x+1)}{(x-1)(x+2)} = \frac{2(x+1)}{x+2}$ for $x \neq 1$.

At $x = -2$ the top is $6 \neq 0$ and the bottom is 0, so $x = -2$ is a vertical asymptote. At $x = 1$ the bad factor cancels and $\lim_{x \rightarrow 1} f(x) = \frac{4}{3}$, a finite number. So the graph has a hole at $(1, \frac{4}{3})$, not an asymptote.

The leading terms are $2x^2$ and x^2 , so $\lim_{x \rightarrow \pm\infty} f(x) = 2$ and $y = 2$ is the horizontal asymptote. ■

6. Continuity and the IVT.

(b) Let $f(x) = \begin{cases} x+2 & x < -1 \\ x^2 & -1 \leq x \leq 1 \\ 3-x & x > 1 \end{cases}$. Is f continuous at $x = -1$ and at $x = 1$? Where is f not differentiable?

Solution. At $x = -1$: $f(-1) = 1$, $\lim_{x \rightarrow -1^-} (x+2) = 1$, $\lim_{x \rightarrow -1^+} x^2 = 1$. All agree, so f is continuous at -1 .

At $x = 1$: $f(1) = 1$ and $\lim_{x \rightarrow 1^-} x^2 = 1$, but $\lim_{x \rightarrow 1^+} (3-x) = 2$. The limit does not exist, so f is not continuous at 1 (a jump).

f is not differentiable at $x = 1$, since a differentiable function is continuous. It is also not differentiable at $x = -1$: the graph is continuous there but has a corner, with slope 1 from the left and slope $2x = -2$ from the right. ■

(d) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.

Solution. Each piece is continuous, so only the seams at $x = 1$ and $x = 4$ matter. The middle line must start at height 2 and end at height 8.

$$a + b = 2, \quad 4a + b = 8.$$

Subtracting, $3a = 6$, so $a = 2$ and $b = 0$. The middle piece is $2x$. ■

(e) Show that $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

Solution. Let $f(x) = x^3 + x - 1$. It is a polynomial, so it is continuous on $[0, 1]$. $f(0) = -1 < 0$ and $f(1) = 1 > 0$.

Since 0 is between $f(0)$ and $f(1)$, the Intermediate Value Theorem gives a number c in $(0, 1)$ with $f(c) = 0$. A continuous graph cannot get from below the axis to above it without crossing. ■

7. The derivative as a limit.

(a) Let $f(x) = x^3$ and $P = (2, 8)$. Find the slope of the tangent line at P as a limit of slopes of secant lines through P and $Q = (x, x^3)$, then write the tangent line.

Solution. The secant through P and Q has slope $\frac{x^3-8}{x-2}$. As $x \rightarrow 2$, Q slides onto P and the secant turns into the tangent. Factor the difference of cubes.

$$m = \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{x-2} = \lim_{x \rightarrow 2} (x^2+2x+4) = 12.$$

The tangent line is $y - 8 = 12(x - 2)$, that is $y = 12x - 16$. ■

(b) Compute $f'(x)$ directly from $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ for $f(x) = \frac{x}{x+2}$.

Solution. Combine the fractions in the numerator first.

$$\frac{1}{h} \left(\frac{x+h}{x+h+2} - \frac{x}{x+2} \right) = \frac{(x+h)(x+2) - x(x+h+2)}{h(x+h+2)(x+2)} = \frac{2h}{h(x+h+2)(x+2)}.$$

Cancel h and let $h \rightarrow 0$: $f'(x) = \frac{2}{(x+2)^2}$. ■

- (d) A ball's height is $h(t) = 200 - 16t^2$ feet after t seconds. Find the slope of the secant line between $t = 1$ and $t = 3$, then the instantaneous velocity at $t = 2$ as a limit.

Solution. $h(1) = 184$ and $h(3) = 56$, so the average velocity is $\frac{56-184}{3-1} = -64$ ft/s.

For the instantaneous velocity at $t = 2$ we use a small time step k (not h , which already names the height).

$$\lim_{k \rightarrow 0} \frac{h(2+k) - h(2)}{k} = \lim_{k \rightarrow 0} \frac{-16(4k + k^2)}{k} = \lim_{k \rightarrow 0} (-64 - 16k) = -64 \text{ ft/s.}$$

The two agree. For a parabola the secant slope over an interval equals the tangent slope at the midpoint. ■

- (e) Show that $f(x) = \sqrt[3]{x}$ is continuous at $x = 0$ but not differentiable there. What does the tangent line at $x = 0$ look like?

Solution. $\lim_{x \rightarrow 0} \sqrt[3]{x} = 0 = f(0)$, so f is continuous at 0. The difference quotient at 0 is

$$\frac{f(0+h) - f(0)}{h} = \frac{h^{1/3}}{h} = \frac{1}{h^{2/3}} \rightarrow \infty \quad (h \rightarrow 0),$$

since $h^{2/3}$ is a small positive number on both sides of 0. The limit is not a number, so $f'(0)$ does not exist. The secant slopes grow without bound, and the tangent line at the origin is vertical: the line $x = 0$. ■

- (f) Each limit is $f'(a)$ for some f and a . Find f and a , then evaluate: $\lim_{h \rightarrow 0} \frac{(2+h)^4 - 16}{h}$ and $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \frac{\pi}{2}}$.

Solution. The first has the shape $\frac{f(a+h)-f(a)}{h}$ with $f(x) = x^4$ and $a = 2$, since $2^4 = 16$. So it equals $f'(2) = 4 \cdot 2^3 = 32$.

The second has the shape $\frac{f(x)-f(a)}{x-a}$ with $f(x) = \cos x$ and $a = \frac{\pi}{2}$, since $\cos \frac{\pi}{2} = 0$. So it equals $f'(\frac{\pi}{2}) = -\sin \frac{\pi}{2} = -1$. ■

8. Differentiation rules.

- (c) Find y' if $y = \frac{x^2+1}{2x+1}$.

Solution. Quotient rule: bottom times derivative of top, minus top times derivative of bottom, over bottom squared.

$$y' = \frac{2x(2x+1) - (x^2+1)(2)}{(2x+1)^2} = \frac{2x^2 + 2x - 2}{(2x+1)^2}.$$

- (d) Find the points on the curve $y = x^3 - 3x^2 - 9x + 2$ where the tangent line is parallel to the line $y = 15x$.

Solution. Parallel lines have equal slopes, so we need $y' = 15$.

$$3x^2 - 6x - 9 = 15 \implies x^2 - 2x - 8 = 0 \implies (x - 4)(x + 2) = 0.$$

At $x = 4$, $y = 64 - 48 - 36 + 2 = -18$, and at $x = -2$, $y = -8 - 12 + 18 + 2 = 0$. The points are $(4, -18)$ and $(-2, 0)$. ■

- (f) A particle has position $s(t) = t^3 - 9t^2 + 24t$ meters at time t seconds. Find $v(t)$ and $a(t)$, with units. When is the particle at rest?

Solution.

$$v(t) = s'(t) = 3t^2 - 18t + 24 \text{ m/s}, \quad a(t) = v'(t) = 6t - 18 \text{ m/s}^2.$$

At rest means $v = 0$: $3(t - 2)(t - 4) = 0$, so $t = 2$ s and $t = 4$ s. Between these times $v < 0$ and the particle moves backward. ■

9. Trig derivatives.

- (a) Starting only from $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$, derive $\frac{d}{dx} \sec x$. Then find $\frac{d^2}{dx^2} \tan x$.

Solution. Write $\sec x = \frac{1}{\cos x}$ and use the quotient rule.

$$\frac{d}{dx} \frac{1}{\cos x} = \frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x.$$

In the same way $\frac{d}{dx} \tan x = \frac{\cos x \cos x + \sin x \sin x}{\cos^2 x} = \sec^2 x$. Then $\sec^2 x = (\sec x)^2$, and the chain rule with the formula above gives

$$\frac{d^2}{dx^2} \tan x = 2 \sec x \cdot \sec x \tan x = 2 \sec^2 x \tan x. \quad \blacksquare$$

- (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$.

Solution.

$$\frac{\sin(4x)}{3x} = \frac{4}{3} \cdot \frac{\sin(4x)}{4x} \longrightarrow \frac{4}{3}.$$

For the second, multiply by $\frac{1 + \cos x}{1 + \cos x}$ and use $1 - \cos^2 x = \sin^2 x$.

$$\frac{1 - \cos^2 x}{x \sin x (1 + \cos x)} = \frac{\sin^2 x}{x \sin x (1 + \cos x)} = \frac{\sin x}{x} \cdot \frac{1}{1 + \cos x} \longrightarrow 1 \cdot \frac{1}{2} = \frac{1}{2}. \quad \blacksquare$$

- (d) Find y' if $y = \frac{\sec x}{1 + \tan x}$.

Solution.

$$y' = \frac{\sec x \tan x (1 + \tan x) - \sec x \sec^2 x}{(1 + \tan x)^2} = \frac{\sec x (\tan x + \tan^2 x - \sec^2 x)}{(1 + \tan x)^2}.$$

Since $\sec^2 x = 1 + \tan^2 x$, the parentheses simplify to $\tan x - 1$, and $y' = \frac{\sec x (\tan x - 1)}{(1 + \tan x)^2}$. ■

10. Chain rule.

(a) Use the table to find $(f \circ g)'(1)$, $(g \circ f)'(1)$, and $(fg)'(2)$.

x	0	1	2	3
$f(x)$	2	3	0	1
$f'(x)$	-1	4	2	5
$g(x)$	1	2	3	0
$g'(x)$	3	-2	1	4

Solution. The chain rule evaluates the outer derivative at the inner value:

$$(f \circ g)'(1) = f'(g(1)) g'(1) = f'(2) \cdot (-2) = 2(-2) = -4.$$

$$(g \circ f)'(1) = g'(f(1)) f'(1) = g'(3) \cdot 4 = 16.$$

$$\text{The product rule gives } (fg)'(2) = f'(2)g(2) + f(2)g'(2) = 2 \cdot 3 + 0 \cdot 1 = 6. \quad \blacksquare$$

(b) Find the tangent line to $y = \tan(\sqrt{x})$ at $x = \frac{\pi^2}{16}$.

Solution. At $x = \frac{\pi^2}{16}$, $\sqrt{x} = \frac{\pi}{4}$ and $y = \tan \frac{\pi}{4} = 1$.

$$y' = \sec^2(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}, \quad y' \left(\frac{\pi^2}{16} \right) = 2 \cdot \frac{1}{2 \cdot \frac{\pi}{4}} = \frac{4}{\pi}.$$

The tangent line is $y = 1 + \frac{4}{\pi} \left(x - \frac{\pi^2}{16} \right)$. ■

(c) Find the tangent line to $y = (x^2 - 3)^5$ at $x = 2$.

Solution. $y(2) = 1^5 = 1$ and $y' = 5(x^2 - 3)^4 \cdot 2x$, so $y'(2) = 5 \cdot 1 \cdot 4 = 20$. The tangent line is $y = 1 + 20(x - 2)$, that is $y = 20x - 39$. ■

(h) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = 2 \sin x + \cos(2x)$ has a horizontal tangent line.

Solution. The chain rule gives $\frac{d}{dx} \cos(2x) = -2 \sin(2x)$. Then use $\sin(2x) = 2 \sin x \cos x$ to factor.

$$y' = 2 \cos x - 2 \sin(2x) = 2 \cos x - 4 \sin x \cos x = 2 \cos x (1 - 2 \sin x).$$

$\cos x = 0$ gives $x = \frac{\pi}{2}, \frac{3\pi}{2}$, and $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}, \frac{5\pi}{6}$. The points are $(\frac{\pi}{6}, \frac{3}{2})$, $(\frac{\pi}{2}, 1)$, $(\frac{5\pi}{6}, \frac{3}{2})$, $(\frac{3\pi}{2}, -3)$. ■

11. Implicit differentiation.

(a) For the ellipse $x^2 + 4y^2 = 16$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, simplified using the equation of the ellipse.

Solution. $2x + 8yy' = 0$, so $y' = -\frac{x}{4y}$. Differentiate again with the quotient rule, remembering that y depends on x .

$$y'' = -\frac{4y - x \cdot 4y'}{16y^2} = -\frac{4y + \frac{x^2}{y}}{16y^2} = -\frac{4y^2 + x^2}{16y^3}.$$

On the ellipse $x^2 + 4y^2 = 16$, so $y'' = -\frac{16}{16y^3} = -\frac{1}{y^3}$. On the top half ($y > 0$) the curve is concave down, and on the bottom half it is concave up. ■

- (b) Find the tangent line to $\sin(x + y) = 2x - 2y$ at (π, π) .

Solution. Check the point: $\sin(2\pi) = 0 = 2\pi - 2\pi$. Differentiate.

$$\cos(x + y)(1 + y') = 2 - 2y'.$$

At the point $\cos(2\pi) = 1$, so $1 + y' = 2 - 2y'$ and $y' = \frac{1}{3}$. The tangent line is $y = \pi + \frac{1}{3}(x - \pi)$. ■

- (c) Find the tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$.

Solution. Check the point: $8 + 64 = 72 = 9(2)(4)$. Differentiate, using the product rule on $9xy$.

$$3x^2 + 3y^2y' = 9y + 9xy' \implies y' = \frac{3y - x^2}{y^2 - 3x}.$$

At $(2, 4)$, $y' = \frac{12-4}{16-6} = \frac{4}{5}$, and the tangent line is $y = 4 + \frac{4}{5}(x - 2)$. ■

- (d) For $y^2 = x^3 - 3x + 2$, find $\frac{dy}{dx}$, the tangent line at $(2, 2)$, and all points with a horizontal tangent line.

Solution. $2yy' = 3x^2 - 3$, so $y' = \frac{3x^2-3}{2y}$. At $(2, 2)$, $y' = \frac{9}{4}$ and the tangent line is $y = 2 + \frac{9}{4}(x - 2)$.

A horizontal tangent needs the top to be 0 and the bottom not 0: $x = \pm 1$ with $y \neq 0$. At $x = 1$, $y^2 = 0$, so that point is excluded (the formula gives $\frac{0}{0}$ there). At $x = -1$, $y^2 = 4$, giving $(-1, 2)$ and $(-1, -2)$. ■

12. Related rates.

- (a) A 13 ft ladder leans against a wall. The bottom slides away at 2 ft/s. How fast is the top sliding down when the bottom is 5 ft from the wall?

Solution. Let x be the distance from the wall to the bottom and y the height of the top. Then $x^2 + y^2 = 169$ at every instant, so $2xx' + 2yy' = 0$.

When $x = 5$, $y = 12$. So $y' = -\frac{xx'}{y} = -\frac{5 \cdot 2}{12} = -\frac{5}{6}$ ft/s. The top slides down at $\frac{5}{6}$ ft/s. ■

- (b) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at 4 m³/min. How fast is the water level rising when the water is 4 m deep?

Solution. Let h be the depth and r the radius of the water surface. Similar triangles give $\frac{r}{h} = \frac{3}{6}$, so $r = \frac{h}{2}$ and

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{12}, \quad \frac{dV}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}.$$

At $h = 4$: $4 = 4\pi \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{1}{\pi}$ m/min. The factor $\frac{\pi h^2}{4}$ is the area of the water surface, which is why the level rises more slowly as the tank fills. ■

- (d) A 6 ft tall person walks away from a 15 ft lamppost at 3 ft/s. How fast is the tip of the person's shadow moving?

Solution. Let x be the person's distance from the post and s the length of the shadow. The tip is at distance $x + s$. The light ray over the person's head makes two similar triangles.

$$\frac{s}{6} = \frac{x+s}{15} \implies 15s = 6x + 6s \implies s = \frac{2x}{3}.$$

The tip is at $x + s = \frac{5x}{3}$, so it moves at $\frac{5}{3} \cdot 3 = 5$ ft/s, whatever the distance. The shadow itself grows at 2 ft/s. ■

- (e) A marble moves along $x^2 + xy + y^2 = 9$. At $(3, 0)$, x is decreasing at 2 ft/s. Is y increasing or decreasing, and how fast?

Solution.

$$2xx' + x'y + xy' + 2yy' = 0.$$

At $(3, 0)$ with $x' = -2$: $2(3)(-2) + 0 + 3y' + 0 = 0$, so $y' = 4$. The y -coordinate is increasing at 4 ft/s. ■

13. Linear approximation.

- (a) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.

Solution. $f(27) = 3$ and $f'(x) = \frac{1}{3}x^{-2/3}$, so $f'(27) = \frac{1}{27}$. Then $L(x) = 3 + \frac{x-27}{27}$ and $L(27.27) = 3 + \frac{0.27}{27} = 3.01$. The true value is 3.00997... ■

- (b) Find the linearization of $f(x) = \sin x$ at $a = \frac{\pi}{6}$. Use it to approximate $\sin\left(\frac{\pi}{6} + 0.02\right)$.

Solution. $f\left(\frac{\pi}{6}\right) = \frac{1}{2}$ and $f'\left(\frac{\pi}{6}\right) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$, so $L(x) = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right)$.

$$\sin\left(\frac{\pi}{6} + 0.02\right) \approx \frac{1}{2} + \frac{\sqrt{3}}{2}(0.02) = \frac{1}{2} + \frac{\sqrt{3}}{100} \approx 0.5173. \quad \blacksquare$$

- (c) The radius of a sphere is measured as 10 cm with a possible error of 0.05 cm. Estimate the maximum error in the computed volume.

Solution. $V = \frac{4}{3}\pi r^3$, so $dV = 4\pi r^2 dr = 4\pi(100)(0.05) = 20\pi \approx 62.8$ cm³.

Here $4\pi r^2$ is the surface area: an error dr in the radius adds a thin shell of that area and thickness dr . ■

14. Extreme values.

- (a) Find the critical numbers of $f(x) = x^{2/3}(x - 5)$. Which come from $f'(x) = 0$ and which from $f'(x)$ undefined?

Solution. Write $f(x) = x^{5/3} - 5x^{2/3}$. Then

$$f'(x) = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3} = \frac{5x - 10}{3x^{1/3}} = \frac{5(x - 2)}{3x^{1/3}}.$$

$f'(2) = 0$, and $f'(0)$ is undefined while $f(0)$ is defined. So the critical numbers are 2 (horizontal tangent) and 0 (a cusp). ■

- (b) Find the absolute max and min of $f(x) = x^4 - 8x^2 + 3$ on $[-1, 3]$.

Solution. $f'(x) = 4x^3 - 16x = 4x(x - 2)(x + 2)$. The critical numbers in $[-1, 3]$ are 0 and 2 (we drop -2).

$$f(-1) = -4, \quad f(0) = 3, \quad f(2) = -13, \quad f(3) = 12.$$

The absolute max is 12 at $x = 3$ and the absolute min is -13 at $x = 2$. ■

- (d) Find the absolute max and min of $f(x) = x\sqrt{4 - x^2}$ on $[-1, 2]$.

Solution. By the product and chain rules,

$$f'(x) = \sqrt{4 - x^2} - \frac{x^2}{\sqrt{4 - x^2}} = \frac{4 - 2x^2}{\sqrt{4 - x^2}}.$$

$f' = 0$ at $x = \pm\sqrt{2}$, and only $\sqrt{2}$ is in $[-1, 2]$. f' is undefined at $x = 2$, an endpoint anyway.

$f(-1) = -\sqrt{3}$, $f(\sqrt{2}) = \sqrt{2} \cdot \sqrt{2} = 2$, $f(2) = 0$. Max 2 at $x = \sqrt{2}$, min $-\sqrt{3}$ at $x = -1$. ■

- (e) Find the absolute max and min of $f(x) = 2\cos x + \sin(2x)$ on $[0, \pi]$.

Solution. Use $\cos(2x) = 1 - 2\sin^2 x$ to write f' in terms of $\sin x$ only.

$$f'(x) = -2\sin x + 2\cos(2x) = -2(2\sin^2 x + \sin x - 1) = -2(2\sin x - 1)(\sin x + 1).$$

On $[0, \pi]$, $\sin x = -1$ never happens, and $\sin x = \frac{1}{2}$ at $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

$$f(0) = 2, \quad f\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2}, \quad f\left(\frac{5\pi}{6}\right) = -\frac{3\sqrt{3}}{2}, \quad f(\pi) = -2.$$

Max $\frac{3\sqrt{3}}{2} \approx 2.60$ at $x = \frac{\pi}{6}$, min $-\frac{3\sqrt{3}}{2}$ at $x = \frac{5\pi}{6}$. ■

15. Mean Value Theorem.

- (a) Find every c guaranteed by the MVT for $f(x) = x^2 - 4x + 1$ on $[1, 5]$.

Solution. $f(1) = -2$ and $f(5) = 6$, so the average slope is $\frac{8}{4} = 2$. Solve $f'(c) = 2c - 4 = 2$: $c = 3$. ■

- (b) If $f(2) = -1$ and $f'(x) \leq 3$ for all x , how large can $f(6)$ be?

Solution. By the MVT on $[2, 6]$, $f(6) - f(2) = f'(c)(6 - 2)$ for some c . Since $f'(c) \leq 3$, $f(6) \leq -1 + 3 \cdot 4 = 11$.

The bound is reached by the line $f(x) = -1 + 3(x - 2)$, so 11 is the best possible. ■

- (d) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

Solution. Let $f(x) = x^3 + 5x - 2$. At least one: $f(0) = -2 < 0 < 4 = f(1)$, so the IVT gives a root in $(0, 1)$.

At most one: if $f(a) = f(b) = 0$ with $a < b$, the MVT gives c with $f'(c) = \frac{0-0}{b-a} = 0$. But $f'(x) = 3x^2 + 5 \geq 5$. So a second root is impossible. ■

16. Shape of a graph.

- (a) For $f(x) = x^3 - 6x^2 + 9x + 1$, find the intervals of increase and decrease, local extrema, concavity, and inflection points.

Solution. $f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$. So $f' > 0$ on $(-\infty, 1)$ and $(3, \infty)$ (increasing) and $f' < 0$ on $(1, 3)$ (decreasing).

f' changes from $+$ to $-$ at 1: local max $f(1) = 5$. From $-$ to $+$ at 3: local min $f(3) = 1$.

$f''(x) = 6x - 12$: concave down on $(-\infty, 2)$, concave up on $(2, \infty)$, inflection point $(2, 3)$. The inflection point sits halfway between the max and min, as for every cubic. ■

- (b) Do the same for $f(x) = 3x^4 - 4x^3$.

Solution. $f'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$. The factor $12x^2 \geq 0$, so the sign of f' is the sign of $x - 1$: decreasing on $(-\infty, 1)$, increasing on $(1, \infty)$.

Local min $f(1) = -1$. The critical number $x = 0$ is not an extremum, since f' does not change sign there. The tangent is flat but the graph keeps falling.

$f''(x) = 36x^2 - 24x = 12x(3x - 2)$: concave up on $(-\infty, 0)$ and $(\frac{2}{3}, \infty)$, concave down on $(0, \frac{2}{3})$. Inflection points $(0, 0)$ and $(\frac{2}{3}, -\frac{16}{27})$. ■

- (f) Find constants a and b so that $f(x) = x^3 + ax^2 + bx$ has an inflection point at $x = 1$ and a local minimum at $x = 3$. Where is its local maximum?

Solution. $f''(x) = 6x + 2a$, and the inflection point needs $f''(1) = 6 + 2a = 0$, so $a = -3$.

$f'(x) = 3x^2 - 6x + b$, and the local min needs $f'(3) = 27 - 18 + b = 0$, so $b = -9$.

Check: $f''(3) = 12 > 0$, so $x = 3$ really is a local min.

Now $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$, and $f''(-1) = -12 < 0$, so the local max is at $x = -1$, with $f(-1) = 5$. The inflection point at 1 sits halfway between -1 and 3, as it must for a cubic. ■

17. Curve sketching.

- (a) Sketch $f(x) = \frac{1}{x^2+3}$, labeling intercepts, asymptotes, extrema, and inflection points.

Solution. Domain $\mathbb{R}$, since $x^2 + 3 > 0$. No x -intercept, y -intercept $(0, \frac{1}{3})$. No vertical asymptote, and $f \rightarrow 0$ as $x \rightarrow \pm\infty$, so $y = 0$ is the horizontal asymptote. f is even.

$$f'(x) = \frac{-2x}{(x^2+3)^2}, \quad f''(x) = \frac{6(x^2-1)}{(x^2+3)^3}.$$

Increasing on $(-\infty, 0)$, decreasing on $(0, \infty)$, local max $(0, \frac{1}{3})$. Concave up on $(-\infty, -1)$ and $(1, \infty)$, concave down on $(-1, 1)$, inflection points $(\pm 1, \frac{1}{4})$. The graph is a bump that flattens toward the x -axis. ■

- (b) Sketch $f(x) = \frac{x^2}{x-1}$, labeling all key features.

Solution. Domain $x \neq 1$. The only intercept is $(0, 0)$. At $x = 1$ the top is $1 \neq 0$, so $x = 1$ is a vertical asymptote: $f \rightarrow -\infty$ as $x \rightarrow 1^-$ and $f \rightarrow \infty$ as $x \rightarrow 1^+$.

There is no horizontal asymptote, since the top has higher degree. Long division gives $\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$, and $\frac{1}{x-1} \rightarrow 0$, so the graph approaches the slant asymptote $y = x + 1$ as $x \rightarrow \pm\infty$.

$$f'(x) = \frac{x(x-2)}{(x-1)^2}, \quad f''(x) = \frac{2}{(x-1)^3}.$$

Increasing on $(-\infty, 0)$ and $(2, \infty)$, decreasing on $(0, 1)$ and $(1, 2)$. Local max $(0, 0)$, local min $(2, 4)$. Concave down on $(-\infty, 1)$, concave up on $(1, \infty)$, no inflection point. The graph is two branches hugging $y = x + 1$, one with a peak at the origin and one with a valley at $(2, 4)$. ■

- (c) Sketch $f(x) = x - 2\sqrt{x}$, labeling all key features.

Solution. Domain $[0, \infty)$. Intercepts: $x - 2\sqrt{x} = \sqrt{x}(\sqrt{x} - 2) = 0$ at $x = 0$ and $x = 4$. No asymptotes.

$$f'(x) = 1 - \frac{1}{\sqrt{x}}, \quad f''(x) = \frac{1}{2x^{3/2}}.$$

$f' < 0$ on $(0, 1)$ and $f' > 0$ on $(1, \infty)$, so the local (and absolute) min is $(1, -1)$. $f'' > 0$, so the graph is concave up everywhere with no inflection point. At $x = 0$ the slope tends to $-\infty$, so the graph leaves the origin heading straight down. ■

18. Optimization.

- (a) A closed box has a rectangular base whose length is twice its width, and its volume is 72 ft^3 . Find the dimensions that use the least material.

Solution. Let the base be w by $2w$ and the height h . The volume gives $2w^2h = 72$, so $h = \frac{36}{w^2}$. The material is the surface area

$$S = 2(2w^2) + 2(wh) + 2(2wh) = 4w^2 + 6wh = 4w^2 + \frac{216}{w}, \quad w > 0.$$

$S'(w) = 8w - \frac{216}{w^2} = 0$ gives $w^3 = 27$, so $w = 3$. $S'' = 8 + \frac{432}{w^3} > 0$, so this is a min. The box is 3 ft by 6 ft by 4 ft, using 108 ft^2 . ■

- (b) A closed cylindrical can holds $16\pi \text{ cm}^3$. Find the radius and height that minimize its surface area.

Solution. $\pi r^2 h = 16\pi$, so $h = \frac{16}{r^2}$. The surface area is two disks plus the side.

$$S(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{32\pi}{r}, \quad S'(r) = 4\pi r - \frac{32\pi}{r^2}.$$

$S' = 0$ gives $r^3 = 8$, so $r = 2 \text{ cm}$ and $h = 4 \text{ cm}$. $S'' > 0$ for $r > 0$, so this is a min. The best can has height equal to its diameter. ■

- (c) A rectangle has its base on the x -axis and its two upper corners on the semicircle $y = \sqrt{25 - x^2}$. Find the largest possible area.

Solution. Put the upper corners at $(\pm x, \sqrt{25 - x^2})$ with $0 \leq x \leq 5$. Then

$$A(x) = 2x\sqrt{25 - x^2}, \quad A'(x) = 2\sqrt{25 - x^2} - \frac{2x^2}{\sqrt{25 - x^2}} = \frac{2(25 - 2x^2)}{\sqrt{25 - x^2}}.$$

$A' = 0$ at $x = \frac{5}{\sqrt{2}}$. Since $A(0) = A(5) = 0$, this is the max, and $A = 2 \cdot \frac{5}{\sqrt{2}} \cdot \frac{5}{\sqrt{2}} = 25$. The best rectangle is twice as wide as it is tall. ■

- (d) A rectangular pen is split in half by a divider parallel to one side. The outer fence costs \$4 per meter and the divider \$2 per meter. With \$200, what is the largest possible area?

Solution. Let the pen be x by y meters with the divider of length y . Cost: $4(2x + 2y) + 2y = 8x + 10y = 200$, so $y = 20 - 0.8x$.

$$A(x) = x(20 - 0.8x) = 20x - 0.8x^2, \quad A'(x) = 20 - 1.6x.$$

$A' = 0$ at $x = 12.5$, and $A'' < 0$. Then $y = 10$, and the largest area is 125 m^2 . ■

- (e) Find the points on the parabola $y = x^2$ closest to the point $(0, 2)$.

Solution. Minimize the squared distance (it has the same minimizer and no root).

$$D(x) = x^2 + (x^2 - 2)^2 = x^4 - 3x^2 + 4, \quad D'(x) = 4x^3 - 6x = 2x(2x^2 - 3).$$

Critical numbers $x = 0$ and $x = \pm \frac{\sqrt{6}}{2}$. $D(0) = 4$ while $D\left(\pm \frac{\sqrt{6}}{2}\right) = \frac{7}{4}$, so the closest points are $\left(\pm \frac{\sqrt{6}}{2}, \frac{3}{2}\right)$, at distance $\frac{\sqrt{7}}{2}$. The bottom of the parabola, $(0, 0)$, is a local max of the distance here, not the min. ■

19. Antiderivatives.

- (b) Find the most general antiderivative of $g(x) = \frac{x^2 - 3x + 2\sqrt{x}}{x}$.

Solution. Split the fraction into powers of x first: $g(x) = x - 3 + 2x^{-1/2}$.

$$G(x) = \frac{x^2}{2} - 3x + 4\sqrt{x} + C. \quad \blacksquare$$

- (d) If $f'(x) = 3\sqrt{x} - 2$ and $f(4) = 10$, find $f(x)$.

Solution. $f(x) = 2x^{3/2} - 2x + C$. Then $f(4) = 16 - 8 + C = 10$ gives $C = 2$, so $f(x) = 2x^{3/2} - 2x + 2$. ■

- (e) A particle has $a(t) = 6t - 4$, $v(0) = 3$, $s(0) = 1$. Find $s(t)$.

Solution. $v(t) = 3t^2 - 4t + 3$ (the constant is $v(0)$), and $s(t) = t^3 - 2t^2 + 3t + 1$ (the constant is $s(0)$). ■

- (f) Let F be the antiderivative of $f(x) = x^2 - 4$ with $F(0) = 0$. Without finding F , decide where F is increasing, where it has local extrema, and where it is concave up. Then find F .

Solution. $F' = f = (x - 2)(x + 2)$, so F increases on $(-\infty, -2)$ and $(2, \infty)$ and decreases on $(-2, 2)$. Local max at $x = -2$, local min at $x = 2$.

$F'' = f' = 2x$, so F is concave up on $(0, \infty)$ and concave down on $(-\infty, 0)$. Where f crosses zero, F has a peak or valley, and where f has its min, F has an inflection point. Finally $F(x) = \frac{x^3}{3} - 4x$, with $F(-2) = \frac{16}{3}$ and $F(2) = -\frac{16}{3}$, which matches the picture. ■

20. Riemann sums.

- (a) Estimate $\int_0^\pi \sin x \, dx$ with the midpoint rule and $n = 3$.

Solution. $\Delta x = \frac{\pi}{3}$, and the midpoints of $[0, \frac{\pi}{3}]$, $[\frac{\pi}{3}, \frac{2\pi}{3}]$, $[\frac{2\pi}{3}, \pi]$ are $\frac{\pi}{6}$, $\frac{\pi}{2}$, $\frac{5\pi}{6}$, with heights $\frac{1}{2}$, 1 , $\frac{1}{2}$.

$$M_3 = \frac{\pi}{3} \left(\frac{1}{2} + 1 + \frac{1}{2} \right) = \frac{2\pi}{3} \approx 2.09.$$

The exact area is 2. The midpoint rule over estimates here because $\sin x$ is concave down on $[0, \pi]$. ■

- (b) Approximate the area under $f(x) = x^2$ on $[0, 4]$ with $n = 4$ using right endpoints, left endpoints, and midpoints. Which are over and which are under?

Solution. $\Delta x = 1$.

$$R_4 = 1 + 4 + 9 + 16 = 30, \quad L_4 = 0 + 1 + 4 + 9 = 14, \quad M_4 = 0.25 + 2.25 + 6.25 + 12.25 = 21.$$

The exact area is $\frac{64}{3} \approx 21.33$. f is increasing, so the right endpoint is the tallest point of each strip and R_4 is an over estimate, while L_4 is an under estimate. M_4 comes out under here because the graph is concave up. ■

- (d) Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

Solution. Read $\Delta x = \frac{3}{n}$ and $x_i = 1 + \frac{3i}{n}$, so the interval starts at 1 and has length 3, and the height is $\sqrt{x_i}$.

$$\int_1^4 \sqrt{x} dx = \frac{2}{3} [x^{3/2}]_1^4 = \frac{2}{3}(8 - 1) = \frac{14}{3}.$$

■

21. The integral as area.

(a) Evaluate $\int_0^2 (3x + \sqrt{4 - x^2}) dx$ with geometry.

Solution. $\int_0^2 3x dx$ is a triangle with base 2 and height 6: area 6. $y = \sqrt{4 - x^2}$ is the top half of the circle of radius 2, so $\int_0^2 \sqrt{4 - x^2} dx$ is a quarter disk: area π .

Total: $6 + \pi$.

■

(b) Evaluate $\int_{-2}^2 (x^3 + \sqrt{4 - x^2}) dx$ with geometry.

Solution. x^3 is odd: its graph on $[-2, 0]$ is the graph on $[0, 2]$ turned upside down, so the area below the axis cancels the area above, and $\int_{-2}^2 x^3 dx = 0$.

$\int_{-2}^2 \sqrt{4 - x^2} dx$ is a half disk of radius 2, area 2π . Total: 2π .

■

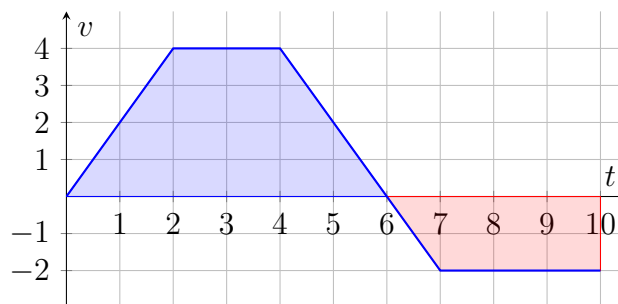
(c) Evaluate $\int_{-3}^3 (|x| - 1) dx$ with geometry.

Solution. The graph is a V with its tip at $(0, -1)$, crossing the axis at $x = \pm 1$. Below the axis is a triangle with base 2 and height 1: area 1. Above the axis are two triangles, each with base 2 and height 2: area 2 each.

$$\int_{-3}^3 (|x| - 1) dx = 4 - 1 = 3.$$

■

(e) For the velocity graph below (v in m/s, t in s), find the displacement and total distance from $t = 0$ to $t = 10$, the position at $t = 10$ if $x(0) = 3$ m, the acceleration at $t = 5$, and when the particle is farthest from its start.



Solution. Displacement is the signed area under v . Above the axis (blue): triangle 4, rectangle 8, triangle 4, total 16. Below (red): triangle 1 and rectangle 6, total 7.

$$\text{displacement} = 16 - 7 = 9 \text{ m}, \quad \text{distance} = 16 + 7 = 23 \text{ m}.$$

The position at $t = 10$ is $3 + 9 = 12$ m. The acceleration at $t = 5$ is the slope of the segment from $(4, 4)$ to $(7, -2)$, which is -2 m/s^2 .

The particle moves forward while $v > 0$ and backward after $t = 6$, so it is farthest from the start at $t = 6$, 16 m away. ■

22. Fundamental Theorem, Part 1.

- (b) Find $g'(x)$ if $g(x) = \int_{x^2}^4 t\sqrt{1+t^2} dt$.

Solution. Flip the limits so the variable is on top: $g(x) = -\int_4^{x^2} t\sqrt{1+t^2} dt$. Then

$$g'(x) = -x^2\sqrt{1+x^4} \cdot 2x = -2x^3\sqrt{1+x^4}.$$

■

- (c) Find $F'(x)$ if $F(x) = \int_0^{\tan x} \sqrt{1+t^2} dt$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and simplify.

Solution. FTC 1 with the chain rule gives $F'(x) = \sqrt{1+\tan^2 x} \cdot \sec^2 x$.

Since $1 + \tan^2 x = \sec^2 x$ and $\sec x > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\sqrt{1+\tan^2 x} = \sec x$, and $F'(x) = \sec^3 x$. ■

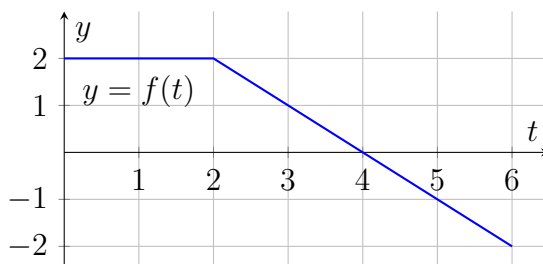
- (d) Find $g'(x)$ if $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$.

Solution. Split at 0 and apply FTC 1 with the chain rule to each piece.

$$g'(x) = \frac{3x^2}{1+x^{12}} - \frac{2}{1+16x^4}.$$

■

- (e) Let $g(x) = \int_0^x f(t)dt$, where f is graphed below. Find $g(4)$, $g'(4)$, $g''(4)$, and where on $[0, 6]$ g is largest.



Solution. $g(4)$ is the area from 0 to 4: a 2×2 square plus a triangle with base 2 and height 2, so $g(4) = 4 + 2 = 6$.

By FTC 1, $g'(4) = f(4) = 0$, and $g''(4) = f'(4) = -1$, the slope of the line.

g increases while $f > 0$ and decreases once $f < 0$, so g is largest at $x = 4$, with $g(4) = 6$. ■

23. Fundamental Theorem, Part 2.

- (a) Evaluate $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2}\right) dx$.

Solution.

$$\int_1^4 (3x^{1/2} - 2x^{-2}) dx = \left[2x^{3/2} + \frac{2}{x}\right]_1^4 = \left(16 + \frac{1}{2}\right) - (2 + 2) = \frac{25}{2}.$$

■

- (c) Evaluate $\int_0^\pi (\sin x + 2 \cos x) dx$.

Solution.

$$[-\cos x + 2 \sin x]_0^\pi = (1 + 0) - (-1 + 0) = 2.$$

The $\cos x$ part contributes 0: its area above the axis on $[0, \frac{\pi}{2}]$ cancels its area below on $[\frac{\pi}{2}, \pi]$.

■

- (d) What is wrong with $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

Solution. FTC 2 needs f continuous on $[a, b]$, and $\frac{1}{x^2}$ blows up at $x = 0$, inside $[-1, 2]$. The answer cannot be right anyway, because $\frac{1}{x^2} > 0$ and the integral of a positive function cannot be negative. (This integral is improper, and it diverges. Improper integrals come in Calculus 2.)

■

24. Substitution.

- (a) Evaluate $\int \sec^2 x \tan^3 x dx$.

Solution. Let $u = \tan x$, so $du = \sec^2 x dx$.

$$\int u^3 du = \frac{u^4}{4} + C = \frac{\tan^4 x}{4} + C.$$

■

- (b) Evaluate $\int_0^{\pi/2} \cos x \sqrt{1 + 3 \sin x} dx$.

Solution. Let $u = 1 + 3 \sin x$, so $du = 3 \cos x dx$ and $\cos x dx = \frac{du}{3}$. The bounds become $u = 1$ and $u = 4$.

$$\frac{1}{3} \int_1^4 u^{1/2} du = \frac{1}{3} \cdot \frac{2}{3} [u^{3/2}]_1^4 = \frac{2}{9} (8 - 1) = \frac{14}{9}.$$

■

- (e) Evaluate $\int_{-2}^1 x \sqrt{x+3} dx$.

Solution. Let $u = x + 3$, so $du = dx$, $x = u - 3$, and the bounds become 1 and 4.

$$\int_1^4 (u - 3)u^{1/2} du = \left[\frac{2}{5}u^{5/2} - 2u^{3/2} \right]_1^4 = \left(\frac{64}{5} - 16 \right) - \left(\frac{2}{5} - 2 \right) = -\frac{8}{5}.$$

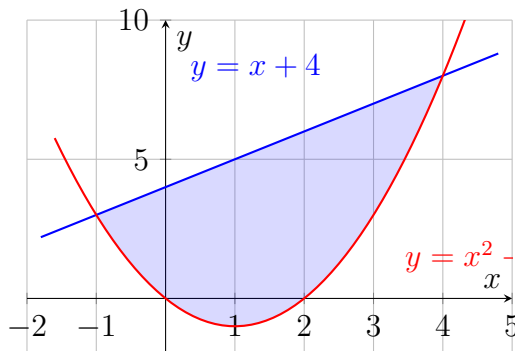
The answer is negative because $x\sqrt{x+3}$ is negative on most of $[-2, 1]$. ■

(f) Evaluate $\int_{-\pi/2}^{\pi/2} x^2 \sin x dx$.

Solution. $f(x) = x^2 \sin x$ is odd: $f(-x) = x^2 \sin(-x) = -f(x)$. On an interval symmetric about 0 the area below the axis on one side cancels the area above on the other, so the integral is 0, with no antiderivative needed. ■

25. Area between curves.

(a) Sketch the region between $y = x^2 - 2x$ and $y = x + 4$ and find its area.



Solution. They meet where $x^2 - 2x = x + 4$, that is $(x - 4)(x + 1) = 0$: $x = -1, 4$. Between them the line is on top (at $x = 0$: $4 > 0$).

$$\int_{-1}^4 (x + 4 - x^2 + 2x) dx = \left[\frac{3x^2}{2} + 4x - \frac{x^3}{3} \right]_{-1}^4 = \frac{56}{3} + \frac{13}{6} = \frac{125}{6}.$$

(b) Find the area between $y = \sqrt{x}$ and $y = \frac{x}{3}$, integrating with respect to x , then with respect to y .

Solution. They meet where $\sqrt{x} = \frac{x}{3}$: $x = 0$ and $x = 9$, at heights 0 and 3. The root is on top (at $x = 1$: $1 > \frac{1}{3}$).

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = \left[\frac{2}{3}x^{3/2} - \frac{x^2}{6} \right]_0^9 = 18 - \frac{27}{2} = \frac{9}{2}.$$

In y , solve each curve for x : the line is $x = 3y$ (on the right) and the root is $x = y^2$ (on the left).

$$\int_0^3 (3y - y^2) dy = \frac{27}{2} - 9 = \frac{9}{2}.$$

- (c) Find the area between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$.

Solution. The curves cross at $x = \frac{\pi}{4}$, so the top curve changes there. Split the integral.

$$\int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2\sqrt{2} - 2.$$

Integrating $\cos x - \sin x$ over the whole interval would give 0, since the two pieces cancel. ■

26. Average value.

- (a) Find the average value of $f(x) = \frac{1}{x^2}$ on $[1, 3]$.

Solution.

$$\frac{1}{3-1} \int_1^3 x^{-2} dx = \frac{1}{2} \left[-\frac{1}{x} \right]_1^3 = \frac{1}{2} \left(1 - \frac{1}{3} \right) = \frac{1}{3}.$$

■

- (b) Find the average value of $f(x) = x^2$ on $[0, 3]$ and a c in $[0, 3]$ with $f(c)$ equal to it.

Solution. $f_{\text{ave}} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \cdot 9 = 3$. Then $c^2 = 3$ gives $c = \sqrt{3}$, which is in $[0, 3]$. The Mean Value Theorem for integrals says such a c always exists for continuous f . ■

27. Volumes.

- (b) Find the volume when the region between $y = x$ and $y = x^2$ is revolved around the x -axis.

Solution. The curves meet at $x = 0, 1$, and $x \geq x^2$ there. A slice at x is a washer with $R = x$, $r = x^2$.

$$V = \pi \int_0^1 (x^2 - x^4) dx = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

■

- (c) Find the volume when the region under $y = \sqrt{x}$ for $0 \leq x \leq 4$ is revolved around the x -axis.

Solution. Each slice is a disk of radius $\sqrt{x}$.

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = 8\pi.$$

■

- (d) Revolve the region between $y = \sqrt{x}$ and $y = x^2$ around the y -axis, and then around the line $x = -1$.

Solution. They meet at $(0, 0)$ and $(1, 1)$. Both axes are vertical, so slice horizontally. At height y the region runs from the parabola $x = y^2$ (inner) out to the root curve $x = \sqrt{y}$ (outer).

$$V_{y\text{-axis}} = \pi \int_0^1 (y - y^4) dy = \pi \left(\frac{1}{2} - \frac{1}{5} \right) = \frac{3\pi}{10}.$$

Around $x = -1$ each radius grows by 1: $R = \sqrt{y} + 1$ and $r = y^2 + 1$.

$$V = \pi \int_0^1 ((\sqrt{y} + 1)^2 - (y^2 + 1)^2) dy = \pi \int_0^1 (y + 2\sqrt{y} - y^4 - 2y^2) dy = \frac{29\pi}{30}.$$

■

28. Work.

- (a) A spring has natural length 20 cm, and 0.9 J of work stretches it from 20 cm to 26 cm. How much work stretches it from 24 cm to 28 cm? What force holds it at a length of 30 cm?

Solution. Work in meters: stretching 0.06 m takes $\int_0^{0.06} kx dx = 0.0018k = 0.9$, so $k = 500$ N/m.

$$W = \int_{0.04}^{0.08} 500x dx = 250(0.0064 - 0.0016) = 1.2 \text{ J}.$$

At 30 cm the stretch is 0.1 m, so the force is $500(0.1) = 50$ N.

■

- (b) A spring has natural length 10 cm and spring constant k . Compare the work W_1 from 10 cm to 15 cm with W_2 from 15 cm to 20 cm.

Solution.

$$W_1 = \int_0^{0.05} kx dx = \frac{k}{800}, \quad W_2 = \int_{0.05}^{0.10} kx dx = \frac{3k}{800}.$$

So $W_2 = 3W_1$. The force grows with the stretch, so the second 5 cm is pulled against a larger force. On the graph of $F = kx$, W_1 is a triangle and W_2 is a trapezoid three times its size.

■

- (c) A 10 m chain with mass 2 kg per meter hangs from the top of a building. How much work does it take to pull the whole chain up to the top? Use $g = 9.8$ m/s².

Solution. After x meters have been pulled up, $10 - x$ meters still hang, with weight $2(9.8)(10 - x) = 19.6(10 - x)$ N. That is the force needed to lift the next small piece dx .

$$W = \int_0^{10} 19.6(10 - x) dx = 19.6 \left[10x - \frac{x^2}{2} \right]_0^{10} = 19.6 \cdot 50 = 980 \text{ J}.$$

Check: the chain's center of mass rises 5 m, and $(20 \text{ kg})(9.8)(5 \text{ m}) = 980$ J.

■

- (d) A force of $F(x) = \frac{x}{(x^2+1)^2}$ pounds acts on an object x feet from the origin. Find the work done moving the object from $x = 0$ to $x = 1$.

Solution. Let $u = x^2 + 1$, so $du = 2x \, dx$, and the bounds become 1 and 2.

$$W = \frac{1}{2} \int_1^2 u^{-2} \, du = \frac{1}{2} \left[-\frac{1}{u} \right]_1^2 = \frac{1}{2} \left(1 - \frac{1}{2} \right) = \frac{1}{4} \text{ ft-lb.}$$

■